



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 2

1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: var is function-scoped, let is block-scoped, and const is used for fixed values.

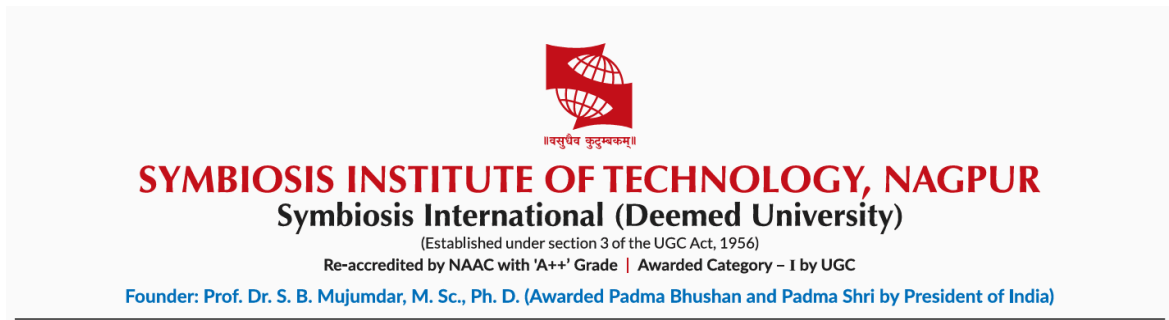
The prompt() function accepts user input, and Number() converts input into numeric values.

The program calculates the total amount, GST, and final bill using arithmetic operations.

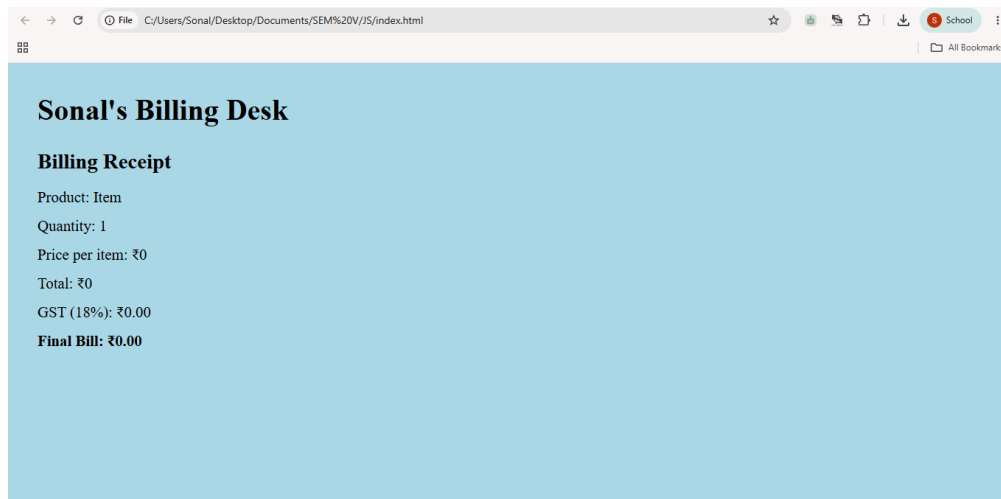
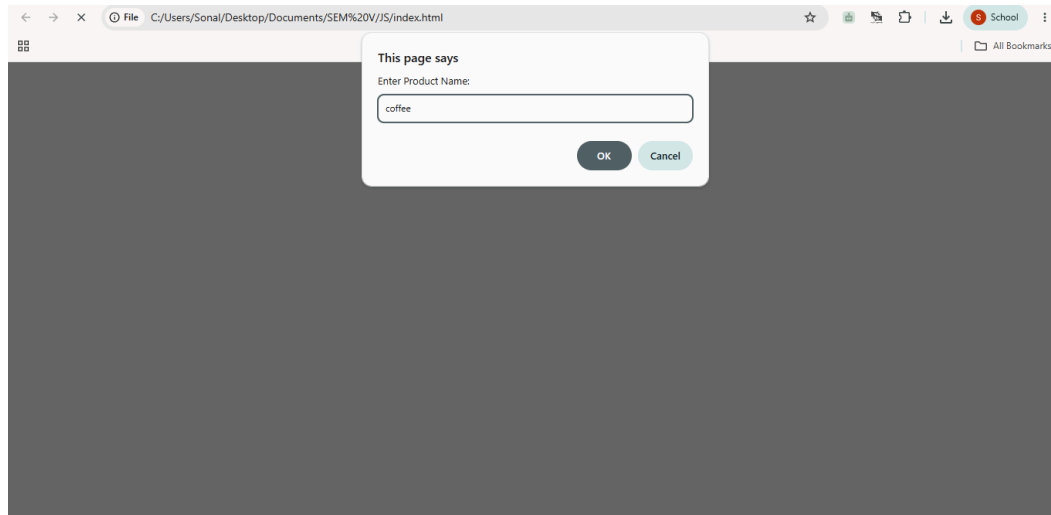
An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using \${}.

The document.write() method displays the billing receipt on the webpage.



4. Output:



5. Result/Conclusion: A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.: 2

1. Case Study Title: Develop the webpage for an online shopping website using HTML, CSS, and HTML tags and calculate the bill, customer name, mobile number, invoice number, product number, quantity (kg), rate per kg, discount, GST, packing charges, payment mode (cash/card/UPI), and membership (yes/no).

2. Case Study Program Code:



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Welcome

Billing.html

Billing.html > html > head

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Online Shopping Bill</title>
5      //Sonal Nakhate
6      //24070521199
7  </head>
8  <body style="background-color: rgba(153, 209, 223, 0.315);">
9      <h1> Online Shopping Website </h1>
10
11     <script>
12         const GST = 0.18;
13
14         // Customer Details
15         var custname = prompt("Enter Customer Name:");
16         var mobile = prompt("Enter Mobile Number:");
17         var invoice = prompt("Enter Invoice Number:");
18         var product = prompt("Enter Product Name:");
19         var quantity = Number(prompt("Enter Quantity (Kg):"));
20         var rate = Number(prompt("Enter Rate per Kg:"));
21         var discount = Number(prompt("Enter Discount (%):"));
22         var packing = Number(prompt("Enter Packing Charges:"));
23         var payment = prompt("Enter Payment Mode (Cash/Card/UPI):");
24         var membership = prompt("Membership (Yes/No):");
25
26         // Calculations
27         let total = quantity * rate;
28         let discountAmt = total * discount / 100;
29         let subtotal = total - discountAmt;
30         let gstAmt = subtotal * GST;
31         let finalBill = subtotal + gstAmt + packing;
```



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Welcome

Billing.html

Billing.html > html > head

```
2   <html>
8   <body style="background-color: rgba(153, 209, 223, 0.315);">
11  <script>
33      // Object
34      const bill = {
35          customer: custname,
36          mobile,
37          invoice,
38          product,
39          quantity,
40          rate,
41          discount,
42          packing,
43          payment,
44          membership,
45          total,
46          discountAmt,
47          gstAmt,
48          finalBill
49      };
50
51      // Destructuring
52      const {
53          customer,
54          mobile: mob,
55          invoice: inv,
56          product: pname,
57          payment: payMode,
58          membership: member,
59          gstAmt: gst,
60          finalBill: amount
```



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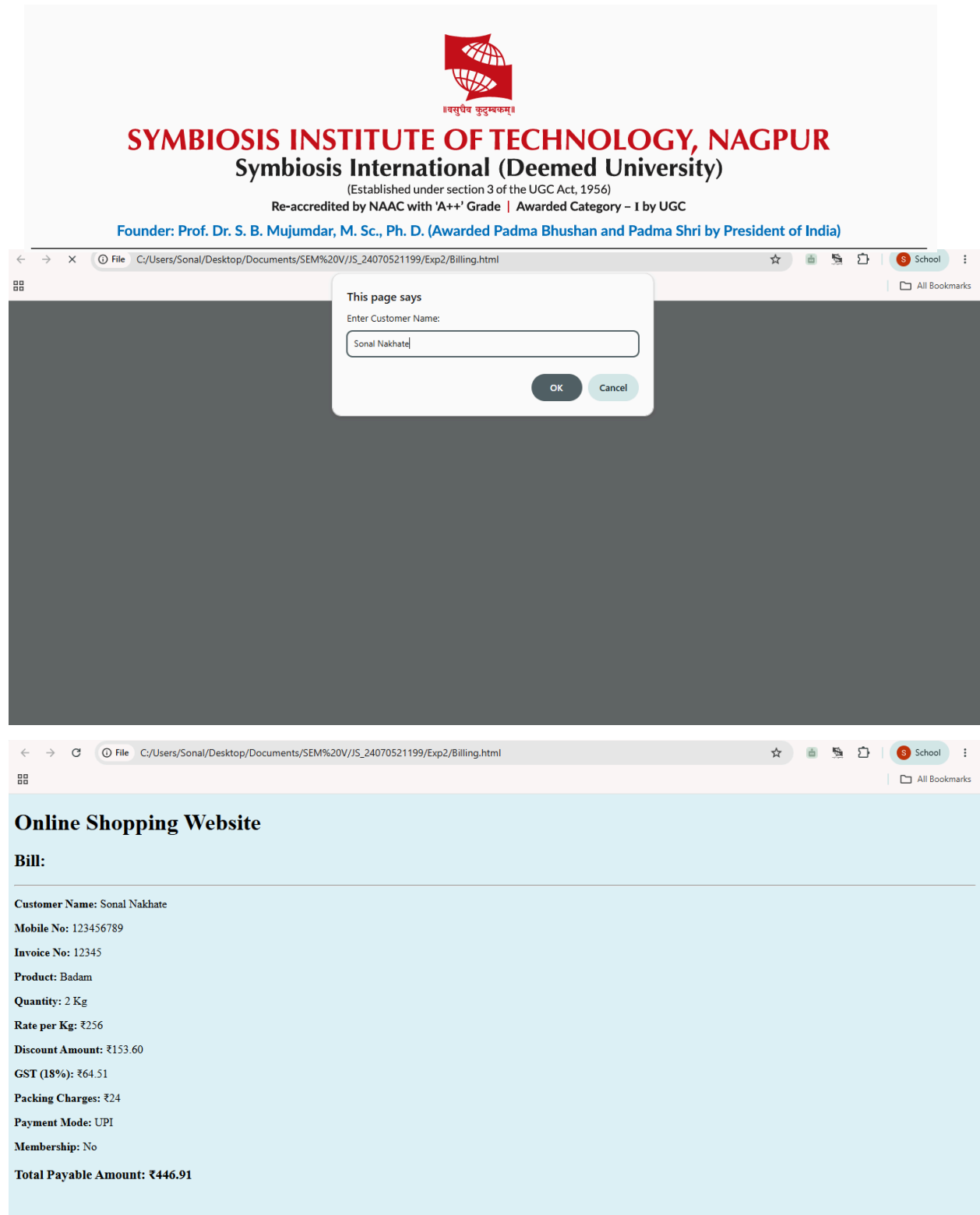
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```
Welcome | <> Billing.html ●
<> Billing.html > html > head
2  <html>
8  <body style="background-color: rgba(153, 209, 223, 0.315);">
11  <script>
58      membership: member,
59      gstAmt: gst,
60      finalBill: amount
61  } = bill;
62
63  // Output
64  document.write(`<h2>Bill: </h2>`);
65  document.write(`<hr>`);
66  document.write(`<p><b>Customer Name:</b> ${customer}</p>`);
67  document.write(`<p><b>Mobile No:</b> ${mob}</p>`);
68  document.write(`<p><b>Invoice No:</b> ${inv}</p>`);
69  document.write(`<p><b>Product:</b> ${pname}</p>`);
70  document.write(`<p><b>Quantity:</b> ${bill.quantity} Kg</p>`);
71  document.write(`<p><b>Rate per Kg:</b> ₹${bill.rate}</p>`);
72  document.write(`<p><b>Discount Amount:</b> ₹${bill.discountAmt.to
73  document.write(`<p><b>GST (18%):</b> ₹${gst.toFixed(2)}</p>`);
74  document.write(`<p><b>Packing Charges:</b> ₹${bill.packing}</p>`);
75  document.write(`<p><b>Payment Mode:</b> ${payMode}</p>`);
76  document.write(`<p><b>Membership:</b> ${member}</p>`);
77  document.write(`<h3><b>Total Payable Amount:</b> ₹${amount.toFixe
78  </script>
79  </body>
80  </html>
```

3. Output:



4. Result/Conclusion: The online shopping billing webpage was successfully developed using HTML, CSS, and JavaScript. The program accepted customer and product details; calculated the



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total bill, including discount, GST, and packing charges; and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.